

Editing Agreement

New requirements from Adler should be in the form of a suggestion. That suggestion will be resolved when our group has updated downstream documents accordingly.

Questions from the Capstone team should be in the form of a comment. Adler or the original commenter should be the ones to resolve that comment once discussion is complete.

Acronyms

- MVP (Minimum Viable Product): minimum features to be developed as part of the capstone project
- Extensions: features not part of the MVP. They might be developed, only if there is time available after the MVP is completed

Project Blurb

Project Title: Collaborative Knowledge Hub for ASC Members

Client: American Society for Cybernetics (ASC)

Contact: adler@asc-cybernetics.org

Project Description

The American Society for Cybernetics (ASC) is seeking a modern Progressive Web App that transforms how its global community connects and collaborates. The platform aims to enable members to create profiles, manage membership and event fees, share research interests, discover collaborators, organize gatherings (online and in-person), and co-create resources like a cybernetic glossary and publication database. The system helps members navigate this ecosystem through knowledge graphs, smart tagging, and thematic connections across disciplines. The interface aims to balance powerful features with intuitive design for a diverse audience.

The Challenge

ASC faces the challenge of managing complex data relationships and diverse content types within a single platform. The project involves designing visualization strategies for knowledge graphs, implementing AI-powered content discovery, and building systems that synthesize information from multiple sources. There is significant diversity in the data, making it essential to identify multiple ways to visualize relationships. The platform aims to harvest online resources, analyze member activity, and synthesize data to enable collaboration among cyberneticians. This could result in automated newsletters featuring cybernetics news, new members, emerging projects, and recent publications.

What We Envision

- **Smart Member Matching:** A recommendation algorithm that connects researchers based on shared interests, complementary projects, and research questions.
- **Event Creation and Coordination:** Tools for organizing online and in-person gatherings with smart scheduling features.
- **Collaborative Knowledge Creation:** A living cybernetic glossary and resource library where members co-create, annotate, link, and curate content.
- **Content Synthesis:** Monthly summaries that analyze member activity, highlight trends, and surface "what's happening in cybernetics."
- **AI-Powered Discovery:** A recommendation engine that searches online for relevant papers, discussions, and potential collaborations based on individual members' profiles.

Why This Matters

This project represents more than a typical web app—it's a platform designed to amplify human collaboration and collective intelligence. The work involves cutting-edge technologies (PWAs, AI/ML, knowledge graphs) while exploring cybernetic principles in practice. The resulting system will serve an international academic community and impact how researchers connect and collaborate.

Requirements Brainstorming

Requirements v0.1 – Concepts

→ Collaborative conversation: your questions and suggestions are welcome. They help to clarify, challenge and improve the requirements, hence the final product.

Questions for Adler

1. What categories of **users** do you want us to support?
 - a. **Admin:** Full system configuration, module creation.
 - b. **Manager:** module specific, e.g., payments, membership, etc.
 - c. **Moderator:** Approves/edits content in the Glossary, Books, and Events modules.
 - d. **Member:** Manages profile, views "Member-Only" content, contributes to modules, content.
 - i. They can have different status (new, active/paid membership dues, dormant, archived account), which defines access rights.
 - e. **Public:** Views public-facing content, e.g., glossary entries and open events.
 - f. NOTE 1: All users (a-d) are likely to be all ASC members, but with different permissions.
 - g. NOTE 2: To account for chapters, e.g., USA, Brazil, Austria, Europe,... there could be a way to group users. A container could be the USA with sub-containers for states or cities. I could create a container called 'Americas' and put USA, Brazil, etc in there. *Maybe* the system doesn't need to understand what a country or state is, just see containers as nested.
2. **Examples of user journeys**
 - a. **Membership lifecycle:** submit interest/application → revision/approval → payment → payment reminders → renew membership → end/archive membership
 - b. **'Playlists' for Organic Learning:** Instead of an endless list of books, articles, or videos, what if it's a journey? It begins with a quest that evolves into a curated list of materials. You can mark what you've read, add reflections or questions. Members could create 'playlists' of materials like "Cybernetics 101", "Orders of Cybernetics", "Cybernetic Organisations".
 - c. **Self-Organised Groups:** What if it were easier to find cyberneticians working on challenges or research questions similar to mine? If desired, these connections could become partnerships in projects, workshops, or writing?
3. What are the **functional requirements**? Which are the most important to you?

OVERVIEW

- a. **Members** (profile with projects/areas/tags, membership info, payment status, preferences - mailing lists, notifications, ...)
- b. **Knowledge hub** (create, manage various types of content: projects, books, glossary, ... and tag content to be able to link / visualise as a node graph)
- c. **Updates** (from the modules, e.g., members joined, new projects, events, etc)
 - i. Content synthesis / summary / AI features
- d. **Events** (create, manage, calendar, event details, export as .ics or add to personal calendars)
- e. **Discovery** (source new content from selected web repositories)

DETAILS

- f. **Membership Tier Management:** Automated dues and status tracking for 9+ tiers (e.g., Student, Lifetime).
 - i. Members tiers: Regular member, Regular (complementary), Student, Affiliate, Equity, Lifetime, Fellow, Emeritus, and some users eventually die or leave the organisation.
 - ii. Non-members: public, guests (non-ASC members who register to events)
 - iii. Manager view: Create, edit, manage, delete, archive user
 - iv. Member view: (edit) profile, follow other members, register to events, mailing lists, etc
 - v. Membership life cycle based on registration, payment dates, etc
 - 1. Add/remove them from google mailing lists using [google APIs like this](#) so that the application you develop communicates with ASC google workspace.
 - 2. Send emails to members (welcome, members dues reminder(s), thanks for renewing membership, status, ...)
 - 3. Process payments
- g. **Admin**
 - i. **Granular Visibility Control:** Toggle to set specific items or modules as 'Public' or 'Members Only'.
 - ii. **Export/Import Engine:** CSV/JSON/iCal tools for data portability and legacy migration.
- h. **Nested Containers:** to group members by regions or chapters (e.g., World > Americas > Brazil). Since a lot happens online, maybe members should be able to follow activities from other regions or groups.
- i. **Events:** manager creates/edits, members (and guests) can register, submit abstracts, review.
 - i. Events index with sorting by years, themes, locations, ...

- ii. Events content, e.g., participants, themes, abstracts with search, sorting and filtering (year, author, theme)
- iii. Event branding: colour scheme, banner ([example](#) of a conference and different [branding](#))
- iv. Similar to events: competitions (cybernetic poster, xmas card, etc) as a way to invite students and a wider audience to participate in cybernetic projects / calls.

j. **Payment Integration:** Native processing for membership renewals and event fees. (*Dependency: Membership Tiers*)

- i. *Providers to consider: Stripe or Zeffy*
- ii. *Types of fees: tiered fees (membership, conferences, events), one-off (donations, some events)*
- iii. *Open questions: have this only or also on the website, but the integration / connection between the website and the application needs to be addressed, e.g., paid member is added to (a) app and (b) google groups*

k. **Knowledge hub**

- i. **Modular Content Creator:** Admin tool to define custom schemas for new repositories (Glossary, Books, etc).
 - 1. Example: conference abstracts, publications with comments, labels/tags, curated content (selection of abstracts by theme, year,...)
- ii. **Member Contribution Workflow:** Interface for members to submit items to modules. (*Dependency: Modular Content Creator*)
- iii. **Moderation Dashboard:** Review queue for managers to approve or edit member submissions. (*Dependency: Member Contribution Workflow*)

l. **Interactive Knowledge Graph:** 2D/3D visualization of relationships between members and tags. (*Dependency: Member Profiles & Module Data*)

m. **Content synthesis**

- i. **AI Metadata Suggestion:** Automated tagging and summarizing of new content submissions. (*Dependency: Moderation Dashboard*)
- ii. **AI Research Harvester:** Scheduled scraping of arXiv and JSTOR for cybernetics-related papers.
- iii. **Automated Newsletter Synthesis:** AI-generated monthly summaries of hub activity for admin review. (*Dependency: AI Harvester & Module Data*)

4. We have assembled a very high level set of 'Draft functional requirements' below.

- 5. What platform+browser combinations are the most important for the PWA to run on?
 - a. OS: Apple, Windows

- b. Browser: Chrome, Safari, Dia (Arc)
- 6. Are there any third parties we need to integrate with?
 - a. Google Workspace
 - b. Payment (Stripe or Zeffy) or alternatively integrate with the website wordpress
- 7. What are the performance requirements
 - a. These [Website Response Times - NN/G](#) are a classic reference
 - b. The app should prioritise stability over tech hype, that is, which modern frameworks will require minimal maintenance or fixing library dependencies?
- 8. How many users are we expecting?
 - a. 250
- 9. Do you have any security requirements/regulations we need to comply with?
 - a. Security: OWASP, 2FA at least for admin, managers, moderators
 - b. Front end and back end should both check for permissions. The front end shouldn't assume that the back-end checks and vice-versa.
 - c. The production data would need to be in the USA
- 10. What can an unauthenticated user view/edit?
 - a. This is feature dependent...
- 11. Do you have any privacy requirements/regulations we need to comply with?
 - a. Privacy: GDPR (for european users), and maybe also California's CCPA/CPRA
- 12. Are there any specific qualities (e.g. portability) that you think are most important for our software to have?
 - a. Maintainability
 - b. Reliability and Scalability, especially with the knowledge library
 - c. Usability
 - d. Modular, easy to add, edit, remove or hide modules
 - i. Import / export functionality in the UI
 - ii. Configurable using environment variables rather than hardcoded paths.
 - e. Interoperability via APIs, if two of these applications are installed, they would be able to communicate with each other, for example, given the right access rights/permissions application A displays members from application B.
 - f. Efficiency (CPU usage)
 - g. Internationalization/Localization
- 13. There needs to be documentation from the early development.

Draft functional requirements:

Extremely high level list of the requirements we identified in the project proposal:

- User management
- Membership fee management
- Events
- Event creation
- Event fee management
- Event coordination
- Knowledge type: cybernetics glossary
- Knowledge type: resource library/publication database
- Collaboration type: Cocreate
- Collaboration Type: Annotate
- Collaboration Type: Link (includes visualization(s) of knowledge graphs)
- Collaboration Type: Curate
- Member matching
- Content synthesis
- AI-Powered Discovery

Curtin Deadlines

Date	Sprint	Producible	Notes
2026-04-05	2	Technology Investigation	SRS 1.0 Requirements
2026-04-26	3	Software Architecture	SRS 1.1 (Requirements)
2026-05-17	4	Handover Plan	SRS 1.2 (Requirements)
2026-05-31	5	Final submission	Freezing requirements

[SRS - Live Version](#)

Meetings

Meetings

[SRS - Live Version](#)

[2026-04-07 Client Meeting](#)

[Summary](#)

[Minutes](#)

2026-04-07 Client Meeting

[Source](#)

Minutes by Jack Davies

Present (online meeting): Jack Davies, Jason Huang, Adler Looks Jorge

Meeting Started 15:03

Meeting Ended 16:13

Next meeting:

- 2026-04-30, 2pm in CET, 8pm in AWST

Summary

- **Potential meeting** with an expert in network analysis
- “Curtin Deadlines” section of the google document should **specify document versions**
- Add bolding and underlining to help make our SRS’ requirements more readable

Requirements:

- PWA support is now an extension, rather than in the MVP. What they want is a responsive web app.
- Events are less important than the Knowledge hub, though still desired.
- **Modular architecture**: it should be possible to wholly replace “Events” with “Events 2.0”
- **MIT Licence** is acceptable, but to prevent patent trolling they want us to upload the code to a repository of theirs every sprint.
- This application (and/or the authentication server we use) should be reusable as an **authentication server** for other projects.
- **Member discussion** and commenting is desired, but need to discuss costs and benefits.

Minutes

Uploaded Version 1.0 to the shared google drive.

Feedback:

- Better changelog, to keep him up to date with new documents. Put the document versions in the "Curtin Deadlines" google doc.
- Stylistically, bold/underline important words. Especially if we have a series of points which are very similar.
- It would be better if we could come up with an agenda for the next meeting

Feedback from the wider ASC:

- Priorities:
 - Progressive web apps are not a requirement. If it's a high development cost, a responsive website would be perfectly acceptable. **PWA support is now an extension, rather than in the MVP.**

Member matching

- The ASC President is looking into developing a tool for this at their university.
- Doesn't need to be very complex, look at keywords, entered and location.
- Adler will think more about what the logic behind this is.

Key Performance Indicator/Northstar

- Members and Knowledge
 - Knowledge = both production of content and consumption of content (playlists, courses)
- **Our key metric is Members completing courses.** It indicates people are actually using the platform and learning. Sharing knowledge with other members.

Knowledge graph

- The point is visualising knowledge.
- "Who is interested in cybernetics and interaction design" + "who is in X area"
- Adler can share examples of this.
 - **Potential meeting with an expert in network analysis**

Licence

- Worried about patent trolls
 - **They want to have a git repository with a copy of the code that gets updated per-sprint, to prove history and licencing**
- MIT Licence should then be acceptable

Authentication

- ASC does not have an authentication server
- They should be able to use this application as an authentication server for other applications

Modular architecture: **it should be possible to wholly replace “Events” with “Events 2.0”**

This app is about 1: building community

- Members making groups with each other
- Hesitant about overstepping on mailing lists
- Node graphs, spontaneous events, member profiles, member matching

This app is also about 2: the knowledge hub

Events

- Have a lot of ways to organise them already. Solutions exist.
- Visualisation would be good
- But it is less important than the knowledge hub, which doesn't yet have a satisfying solution

Server hosting

- We should look into free credit options

Members chatting might be desired

- Weigh efforts and benefits
-

2026-05-22 Client Meeting

We have just finished version 1.2 of our Software Requirements Specification. A link for you to comment on the live version of our SRS:

https://docs.google.com/document/d/1SgeeLRvDJF61Q0Ocn4nX6y-QAlpn3da9UICbDWHr_zY/e
[dit](#). A changelog can be found at the start of the document.

For our meeting this week, some topics we wish to discuss:

1. The most critical section we need to confirm with you is **Appendix B: Handover Plan** of the SRS.

- Our current assumption is that we are not handling installation.
- We have presented a list of VPS solutions that should work if we are conscious about resource usage during development. Is an ~USD\$10 per month price point acceptable?
- During development, we may find that certain features require increased resources - I can imagine many approaches to knowledge graph calculations will be memory intensive. How problematic would an increase to that USD\$10 price point be?

2. The prioritisation of features has had an overhaul. See [Appendix A](#) of the SRS for a full list of their relative priority - you should be able to CTRL-F any of the names there to find the relevant sections.

- Are you satisfied with the new MVP definitions of the Knowledge Graph, Member Matching and Automated Newsletter?
- Are there any priorities you want changed?

3. As always, we would appreciate any feedback on the functional requirements - particularly, we've added in user journeys and feedback mechanisms to the SRS.

The resources you passed on to us from Krebs were very useful, thank you. As requested, a summary of our current understanding of our problem for him:

Our dataset has two main entity types: members and resources (events, books, courses, glossary entries, etc.). All entities have tags, so a simple method of analysis is "number of matching tags". We additionally have a lot of information on interaction between members and resources, including: comments, rating, bookmarks and "has read" - this seems well suited to the two-mode social analysis you describe in 'our Choices Reveal Who You Are: Mining and Visualizing Social Patterns'.

Our challenge is to transform this dataset into two outputs:

- Member matching: make predictions on which members are likely to be interested in collaborating.
- Visualisation: producing a visualisation that allows members to explore relationships between members and concepts (tags).

Additionally, this analysis needs to be updated frequently, and happens on a machine with limited resources and that is running other applications. As such, compute and memory usage are notable constraints on the viable solution space.

Kind regards,

Jack Davies

she/her

22203221

References

- <https://www.dubberly.com/concept-maps/creating-concept-maps.html>
-

Modules

OVERVIEW

1. **Members** (profile with projects/areas/tags, membership info, payment status, preferences - mailing lists, notifications, ...)
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 - a. Content synthesis / summary / AI features
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5. **Discovery** (source new content from selected web repositories)

Administration

Admin

1. **Granular Visibility Control:** Toggle to set specific items or modules as 'Public' or 'Members Only'.
2. **Export/Import Engine:** CSV/JSON/iCal tools for data portability and legacy migration.

User roles

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 - a. They can have different status (new, active/paid membership dues, dormant, archived account), which defines access rights.
5. **Public:** Views public-facing content, e.g., glossary entries and open events.

NOTE 1: All users (1-4) are likely to be all ASC members, but with different permissions.

NOTE 2: To account for chapters, e.g., USA, Brazil, Austria, Europe,... there could be a way to group users. A container could be the USA with sub-containers for states or cities. I could create a container called 'Americas' and put USA, Brazil, etc in there. *Maybe* the system doesn't need to understand what a country or state is, just see containers as nested.

Members

Membership Tier Management: Automated dues and status tracking for 9+ tiers (e.g., Student, Lifetime).

1. Members tiers: Regular member, Regular (complementary), Student, Affiliate, Equity, Lifetime, Fellow, Emeritus, and some users eventually die or leave the organisation.
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 - b. Send emails to members (welcome, members dues reminder(s), thanks for renewing membership, status, ...)
 - c. Process payments

Groups

Nested Containers: to group members by regions or chapters (e.g., World > Americas > Brazil). Since a lot happens online, maybe members should be able to follow activities from other regions or groups.

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Events: manager creates/edits, members (and guests) can register, submit abstracts, review.

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4. Similar to events: competitions (cybernetic poster, xmas card, etc) as a way to invite students and a wider audience to participate in cybernetic projects / calls.

Knowledge Hub

Knowledge hub

1. **Modular Content Creator:** Admin tool to define custom schemas for new repositories (Glossary, Books, etc).
 - a. Example: conference abstracts, publications with comments, labels/tags, curated content (selection of abstracts by theme, year,...)
2. **Member Contribution Workflow:** Interface for members to submit items to modules.
(*Dependency: Modular Content Creator*)
3. **Moderation Dashboard:** Review queue for managers to approve or edit member submissions. (*Dependency: Member Contribution Workflow*)

Knowledge Graph Visualisations

Interactive Knowledge Graph: 2D/3D visualization of relationships between members and tags. (*Dependency: Member Profiles & Module Data*)

Content Synthesis

Content synthesis

1. **AI Metadata Suggestion:** Automated tagging and summarizing of new content submissions. (*Dependency: Moderation Dashboard*)
2. **AI Research Harvester:** Scheduled scraping of arXiv and JSTOR for cybernetics-related papers.
3. **Automated Newsletter Synthesis:** AI-generated monthly summaries of hub activity for admin review. (*Dependency: AI Harvester & Module Data*)

Learning Journeys

Provisional:

- members can submit sequences of knowledge hub entries.

Journeys go through the standard approval process.

- Members can mark which parts of a journey they have completed
 - What if the same entry is on multiple journeys? Is the requirement actually that they can mark any entry as “read”.
- Members can see a list of journeys that they have started/completed/favourited.
- Presumably journeys have some kind of content of their own - e.g. an explanation of how each element fits together, or a particular lens to read each entry through.

Comments

Provisional

This is a low priority requirement “might be desired” “Weigh efforts and benefits”

- Members can leave comments on knowledge hub entries after they have been approved.
- Presumably there should be some mechanism for evaluating the quality and relevance of comments.
- There would need to be a mechanism for reporting harmful comments
- Moderators would need to be able to delete comments, and there would need to be an audit trail for this.

This seems like a moderation burden. Perhaps the same feature of “see how a particular book/article is viewed by the community” is better covered by the learning journeys, as those have an editorial process.

Nonfunctional Requirements

Privacy:

- GDPR (for european users)
- maybe also California's CCPA/CPRA

Security:

- OWASP
- 2FA at least for admin, managers, moderators
- Front end and back end should both check for permissions. The front end shouldn't assume that the back-end checks and vice-versa.
- The production data would need to be in the USA

Performance

- These [Website Response Times - NN/G](#) are a classic reference
- The app should prioritise stability over tech hype, that is, which modern frameworks will require minimal maintenance or fixing library dependencies?
- ~250 users

Clientside interfaces:

- OS: Apple, Windows
- Browser: Chrome, Safari, Dia (Arc)

Additional qualities:

1. Maintainability
2. Reliability and Scalability, especially with the knowledge library
3. Usability
4. Modular, easy to add, edit, remove or hide modules
 - a. Import / export functionality in the UI
 - b. Configurable using environment variables rather than hardcoded paths.
5. Interoperability via APIs, if two of these applications are installed, they would be able to communicate with each other, for example, given the right access rights/permissions application A displays members from application B.
6. Efficiency (CPU usage)
7. Internationalization/Localization
8. There needs to be documentation from the early development.

SRS 1

SRS 1.1

Software Requirement Specification

Version 1.1 | Prepared on X

Previous version: CKH SRS 1.0.pdf

For the American Society of Cybernetic's Collaborative Knowledge Hub By Group 2 of Curtin's
Capstone Computing Project 1

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Changelog

As per 2026-04-07 Client Meeting

- Events management is no longer in the MVP
- The Knowledge Hub is now in the MVP, rather than an extension
- PWA support is no longer in the MVP (they just want a responsive web app) - 4.3
Software interfaces now includes the fact that authentication should be reusable for future ASC endeavours.
- Editing to remove promises about specific implementation technologies. Sections 2.4, 2.5 and 2.6 remain untouched (other than a note that we make no promise about use of specific technologies) as the technologies we intend to use are relevant to the expected environment and constraints of this software.
- Slight revisions to the submission/publication processes in 3.3 Knowledge Hub and 3.5 Event Creation and Coordination.
- New and/or revised requirements for granting of moderator/manager permissions in 3.3 Knowledge Hub, 3.4 Administration and 3.5 Event Creation and Coordination. - Link 3.2 Groups to the ASC google workspace mailing lists.

1. Intro

1.1 Project Description

The Collaborative Knowledge Hub (CKH) is envisioned as a new, standalone responsive web app that integrates multiple systems traditionally found across separate platforms - membership management, event coordination, research discovery, and collaborative knowledge creation - into a single cohesive ecosystem. It will act as both a community portal and a knowledge infrastructure for the American Society for Cybernetics (ASC).

The platform must support complex, interconnected data structures, including relationships between resources such as members, member-produced articles, external publications, glossary terms, and events. Manual and automated metadata tagging will serve as the backbone for producing a knowledge graph, in order to produce visualizations of these relationships and enable intuitive navigation.

The system also incorporates AI-powered features, such as automated content synthesis, smart recommendations, and external resource harvesting, making it more than a traditional membership portal. It is designed to evolve over time as members contribute content, annotate resources, and engage in collaborative activities.

1.2 Definitions

CKH: The Collaborative Knowledge Hub (The product this document describes).

[Extension: <Feature>]: sections with this tag aren't in the MVP.

1.2.1 User Roles

See [2.3 User Characteristics](#) for more detail

Member: an ASC member.

Moderator: a member with some permissions for a particular module, usually for moderating content on the site. These permissions are per-module, and some modules may define more granular permissions (e.g. someone can be a moderator of all events, or just some specific events).

Manager: a member with significant permissions, with near complete control of some aspects of the site. These permissions are per-module, and some modules may define more granular permissions (e.g. someone can be the manager of all events, or just some specific events).

Administrator: a member with permission to configure the system.

2. Overall Description

2.1 Product Perspective

The ASC Knowledge Hub is a standalone web-based platform designed for the American Society for Cybernetics (ASC). Its purpose is to centralise ASC publication, glossary terms, member-generated resources, and cybernetics-related knowledge into a single, searchable, and extensible system.

The system follows a three-tier architecture consisting of:

- Front-end client: A reactive, browser-based interface responsible for browsing, searching, editing, and interacting with ASC resources
- middleware/API layer handles business logic, authentication, authorisation, OTP-based security, and communication between the client and database
- Database layer - storing structured metadata, glossary terms, publications, user accounts, and relationships between concepts.

This architecture supports modularity, long-term extensibility, and the addition of new knowledge modules as ASC evolves. The knowledge Hub will be accessible through modern web browsers on both desktop and mobile devices.

2.2 Product Functions

The platform will include a wide range of high-level capabilities:

Member management

- Profile creation with fields for research interests, affiliations, publications, and personal statements.
- Membership status tracking, including renewal reminders and fee processing.
- Secure authentication and role-based access control.

Collaborative knowledge creation

- A living glossary of cybernetic terms curated by members
- A shared publication and resource library with annotation and linking features
- Version control for glossary entries and collaborative documents

[Extension: Events] Event creation and coordination

- Tools for creating online or in-person gatherings
- Event pages with RSVP tracking, reminders.
- Integrated knowledge creation functionalities for content (e.g. abstracts) submission and review.

[\[Extension: Knowledge Graph\]](#) Knowledge graph visualization

- Interactive graphs showing relationships between resources such as members, member-written articles, external publications, and events.
- Multiple visualization modes to accommodate different cognitive styles.
- Filters, clustering and semantic tagging for deeper exploration.

[\[Extension: Content Synthesis\]](#) AI powered discovery and content synthesis

- Automated monthly summaries in highlighting trends, new members, active discussions, and emerging research
- AI tools that search online for relevant papers, discussions, and events.
- Personalized recommendations based on user profiles and activity

[\[Extension: Content Synthesis\]](#) Smart member matching

- AI-driven algorithms that analyze member profiles, interests, and activity patterns.
- Suggestions for potential collaborators, mentors, or project partners.
- Dynamic matching that updates as members contribute new content

2.3 User Characteristics

The system will be used by:

- General public users interested in cybernetics
 - Individuals interested in cybernetics, system theory, or ASC publications
 - Varying levels of digital literacy
 - Primarily consumers of content rather than contributors
- ASC members (academics, research, practitioners)
 - Typically familiar with the academic platforms, digital libraries, and research tools
 - May contribute new resources, glossary terms, or publications
 - Require reliable search, citation-friendly metadata, and stable access to archival materials
 - Expected to use the platform on both desktop and [\[Extension: Platform Support\]](#) mobile devices
- Moderators
 - Moderates a specific module, or something more granular as defined per-module.
 - Responsible for content moderation, metadata curation, and maintenance.
- Managers
 - Manages a specific module, or something more granular as defined per-module.
 - Responsible for event creation, payments review, publication updates, and membership management.
 - Expected to perform task such as:
 - Managing user roles
 - Auditing payments and membership statuses

- Creating events
- Administrators
 - Require elevated permissions and access to administrative dashboards
 - Expected to perform task such as:
 - Configuring modules
 - Performing data imports and exports
 - Managing user roles
 - May require onboarding to understand metadata structures and editorial workflows

Users are expected to have general digital literacy but do not require technical expertise. Administrators may require brief onboarding to use the content management features.

2.4 Operating Environment

The ASC Knowledge Hub operates within a dedicated server, possibly a Virtual Private Server (VPS). The system is designed to be cost-efficient for the ASC, a non-profit organisation, while still supporting modern web application standards, containerised deployment, secure authentication, and long-term maintainability. This section describes the full technical environment required for the system to function, including client-side, server-side, database, hosting, deployment, network, hardware, and AI-related requirements.

Note that technologies listed in this section are not final: this is our intended stack, and so influence the operating environment, but elements may change during development.

2.4.1 Hardware Environment

Client-side hardware

- The system is designed to run on standard consumer devices with no specialised hardware requirements
 - Desktop or laptop computer
 - Smartphone or tablets
 - Minimum 4 GB RAM recommended for smooth browser performance
 - Minimum display resolution: 1280x720
 - Recommended resolution: 1920x1080
 - Touchscreen support for mobile and tablet users

Server-side hardware

- The ASC Knowledge Hub is deployed on a standalone Linux server. This hosting model was selected because it is significantly more cost-effective for the ASC, a non-profit organisation.
 - Minimum Server specification:
 - 2 vCPUs
 - 4 GB RAM
 - 40 GB SSD storage

- Recommended product specification:
 - 4 vCPUs
 - 8 GB RAM
 - 80+ GB SSD
 - SSD storage for fast I/O
 - off-server backup storage
- The server must support containerised workloads and persistent storage for PostgreSQL.

2.4.2 Software Environment

Client-side software

- React-generated javascript, css and html
- Client-side requirements
 - Modern browser with ES6 support
 - JavaScript enabled
 - Cookies enables for session handling
 - CSS3 support for responsive layouts

Server-side software

- Operating system
 - Linux
- FastAPI server
 - FastAPI
 - Asynchronous request handling
 - Pydantic for schema validation
 - Dependency injection for modularity
- Next.js server
 - React
 - Selected for:
 - Component-based architecture
 - Strong ecosystem and long-term maintainability
 - Excellent documentation and community support
 - Compatibility with REST/JSON APIs
 - High performance through virtual DOM rendering
 - Supports responsive UI design for desktop and mobile devices
 - Integrates with modern build tools
- OJS Application
- Containerisation
 - Docker Compose for orchestrating multi-service deployments
 - Podman as a rootless alternative if preferred or, for some reason, required
- Reverse Proxy
 - Nginx for:
 - HTTPS termination
 - TLS certificate management

- Routing to Next.js, FastAPI or OJS server
- Static file delivery
- Rate limiting

Database

- PostgreSQL is used as the primary storage technology
- Chosen due to its strong relational modelling, JSONB support, ACID compliance, and sustainability for metadata-rich systems
- Persistent volumes for data durability.

Authentication and Security libraries

- Argon2-cffi for secure password hashing
- PyOTP for optional two-factor authentication
- JWT/session-based authentication
- Strict CORS and input validation
- Role-based access control

Backend development tools

- pytest
- uv
- Ruff
- MyPy
- Testing responsibilities include:
 - AI endpoints
 - Database operations
 - Authentication flows
 - Error handling

Front-End development tools

- React testing library (RTL) for behaviour-driven component testing
- Jest for unit tests, snapshot tests, and mocking
- Testing responsibilities include:
 - Component rendering
 - User interaction simulation
 - API call mocking
 - Regression testing for UI behaviour

2.4.3 Network Environment

Connectivity Requirements

- Stable internet connection required for all users
- Low latency recommended for administrative tasks

Protocols

- HTTPS required for all communication
- TLS 1.2+ encryption
- REST/JSON API communication

API Security

- Rate limiting via Nginx
- CORS restricted to ASC domains
- Input sanitisation at API layer

Internal networking

- Docker internal network exposes:
 - FastAPI backend
 - PostgreSQL database
 - Next.js front end
 - OJS application
 - Reverse proxy
- The database is not exposed publicly.

2.4.4 Hosting Environment

Hosting characteristics

- Single Server instance
- No auto-scaling or cloud orchestration
- Full control over environment configuration
- SSH access restricted to authorised administrators

Deployment model

- Containerised using Docker for consistent development and deployment
- Services include:
 - FastAPI server (run with the default server, uvicorn)
 - OJS application
 - Next.js server
 - PostgreSQL database
 - Nginx reverse proxy
- Secrets stored outside version control

Backup strategy

- Snapshot-based backups
 - The server filesystem and PostgreSQL data volumes are periodically captured using snapshot-based backups
 - Snapshots provide point-in-time recovery with minimal downtime
 - Snapshots are stored in external object storage (e.g., S3-compatible storage or low-cost object storage provider)
 - Object storage is chosen because it is:
 - Highly durable
 - Extremely cost-effective
 - Independent of the Server
 - Suitable for long-term archival

- Off-Server storage requirement
 - All backups must be stored outside the server to avoid single-point-of-failure scenarios
 - Object storage ensures that backups remain available even if the server becomes corrupted or inaccessible
- Recovery procedures
 - Snapshots restoration for full-system recovery
 - Logical backup restoration for partial recovery

2.4.5 AI Integration environment

Default AI

- Claude used for:
 - Content summarisation
 - Metadata suggestions
 - Internal editorial tools
- Code will be structured to allow easy transition to other models, but they will not be supported

AI Hosting

- All AI processing occurs is external APIs
- No local model hosting required
- No GPU or specialised hardware required.

2.4.6 External Integrations

Peer-review systems

- Integration with OJS (open Journal Systems)
- Supports:
 - Submission workflows
 - Reviewer assignments
 - Metadata synchronisation

Payment platform

- Stripe for:
 - Membership payments
 - Donations
 - Event fees
- PCI compliance handled by Stripe

2.5 Design and Implementation Constraints

The ASC Knowledge Hub must operate within several technical, organisational, and environmental constraints that influence system design, implementation, and deployment.

Note that technologies listed in this section are not final: this is our intended stack, and so create relevant constraints, but elements may change during development.

2.5.1 Technical constraints

Backend framework constraints

- The backend will be implemented using FastAPI, which enforces asynchronous request handling, Pydantic-based validation, and ASGI-compatible deployment.
- The back end will run on an ASGI server, which affects concurrency models and middleware compatibility.

Front-end framework constraints

- The front-end will be built using React, requiring a component-based architecture, JSX syntax, and a modern JavaScript toolchain
- UI testing will use React Testing Library (RTL) and/or Jest, influencing how components are structured and how side-effects are handled.

Database constraints

- The system will use PostgreSQL as the primary database
- Data modeling must thus be defined with relational structures and JSONB fields
- PostgreSQL will run inside a container with persistent volumes, limiting direct OS-level tuning

Containerisation Constraints

- Services will be deployed using Docker Compose, which requires root access. This is possibly circumventable by using Podman as an intermediary.

Hosting constraints

- The system will run on a single server
- Resource usage must remain within the server limits (CPU, RAM, storage)
- No auto-scaling or distributed deployment is available

2.5.2 Security Constraints

- Passwords must be hashed using Argon2, requiring CPU-intensive hashing and secure configuration.
- We are using PyOTP for two-factor authentication, requiring time-synchronised OTP generation.
- All communication must occur over HTTPS.
- Strict input validation is required to prevent SQL injection, XSS, and CSRF
- Role-based access control must be implemented for administrative features
- Handling of payment information must be PCI-compliant.

2.5.3 Data Protection and Backup Constraints

- PostgreSQL data must be backed up using snapshot-based backups stored in external object storage
- Backups must be stored off-server to avoid single-point-of-failure scenarios

2.5.4 Organisational constraints

- ASC is a non-profit organisation with low revenue, requiring cost-efficient hosting and minimal recurring expenses
- The system must be maintainable by volunteers or part-time administrators
- Technology choices must prioritise long-term sustainability and low operational overhead

2.5.5 Integration constraints

- The system will integrate with OJS for peer-review workflows, requiring compatibility with OJS metadata formats and APIs.
- Stripe will be used for payments, requiring secure API communication
- Claude AI will be the default AI integration.

2.6 Assumptions and Dependencies

The design and operation of the ASC Knowledge Hub rely on several assumptions and external dependencies.

Note that technologies listed in this section are not final: this is our intended stack and our current assumptions, but elements may change during development.

2.6.1 System assumptions

- Users have access to a stable internet connection and a modern web browser
- Administrators possess basic digital literacy and can manage content through the admin interface
- ASC will maintain the accuracy and relevance of glossary terms, publications, and metadata
- The hosting provider will maintain uptime, hardware reliability, and network connectivity
- The system will not require horizontal scaling due to the ASC's modest traffic expectations.

2.6.2 Technical Assumptions

- We have sufficient access to the server to run Docker as root.
- PostgreSQL will remain stable and compatible with FastAPI ORM layers
- Snapshot-based backups and object storage will remain accessible and affordable
- External AI APIs (Claude) will remain available and stable

- Nginx will handle TLS certificates and HTTPS termination reliably.

2.6.3 External dependencies

- Third-party services
 - Stripe must remain operational for payment processing
 - Claude AI must remain available for AI-assisted features
- Software dependencies
 - FastAPI, React, PostgreSQL, Docker, and related libraries must remain maintained
 - OJS must remain maintained for peer-review integration
 - Security libraries (argon2-cffi, PyOTP) must continue receiving updates
 - Testing frameworks (pytest, RTL, Jest) must remain compatible with the codebase.
- Hosting dependencies
 - The server provider must supply CPU, RAM, and storage
 - Server storage must remain available for snapshot-based backups
 - SSH access must remain available for administrative maintenance.

3. Functional Requirements

3.1 Member Management

A person who is a member has a more descriptive identity, with details and permissions specific to their relevant role in ASC Collaborative Knowledge Hub.

Profile Management

REQ-1.1 Members must be able to create a profile.

REQ-1.2 Members must be able to include fields for core profile information/identification, research interests, affiliations, publications and personal statements when creating a profile.

REQ-1.3 Members must be able to link resources to be displayed on their profile.

Membership Status

REQ-1.4 Members must be able to view their membership status.

REQ-1.5 Members must be able to update their membership status, such as cancel their membership or change their membership type.

REQ-1.6 Members must be able to receive reminders (email & push-notification) on their membership status, such as when their membership is soon to expire.

REQ-1.7 Members must be able to view their membership's payment plan.

REQ-1.8 Members must be able to update/change their membership's payment plan.

REQ-1.9 Members must be able to access a payment portal from their membership status page.

Member Moderation

REQ-1.10 Managers must be able to **view** a member's profile.

REQ-1.11 Managers must be able to **create** a member's profile.

REQ-1.12 Managers must be able to **delete** a member's profile.

REQ-1.13 Managers must be able to **edit** a member's profile.

REQ-1.14 Managers must be able to **update** a member's profile details.

REQ-1.15 Managers must be able to **view an audit log** of changes to a member's profile.

REQ-1.16 Managers must be able to view a member **analytics dashboard** for membership data, including renewal rates and pending renewal.

3.2 Groups

REQ-2.1 Members are able to join groups corresponding with the ASC's google workspace mailing lists.

REQ-2.2 Managers are able to assign or unassign users to groups.

3.3 Knowledge Hub

REQ-3.1 Administrators are able to define custom schemas for knowledge hub repositories.

REQ-3.2 Administrators are able to configure repositories as "public" or "members only".

REQ-3.3 Managers are able to configure items as "public" or "members only"

REQ-3.4 Members are able to make submissions to repositories.

REQ-3.5 Moderators are able to request revisions to a submission pre-publication.

REQ-3.6 Members are able to revise a submission pre-publication.

REQ-3.7 Moderators are able to publish a submission.

REQ-3.8 Moderators are able to access the revision history of a submission.

REQ-3.9 Managers are able to give members moderation permissions for a repository.

3.4 Administration

REQ-4.1 Administrators are able to enable/disable non-core modules

REQ-4.2 Administrators are able to set visibility permissions on non-core modules (public-facing, member-only, moderator-only, etc.)

REQ-4.3 Administrators are able to export each module's data as a [TBD] file.

REQ-4.4 Administrators are able to import a [TBD] file to override a module's current data.

REQ-4.5 Managers for a module are able to assign a member to be a moderator for that module.

REQ-4.6 Administrators are able to assign a member to be a manager for a module. N.B. per-module administration powers are included in that module's functional requirements.

3.5 Event Creation and Coordination

[\[Extension: Events\]](#)

An event represents an organised meeting hosted by a manager, providing a central information panel with event details and engagement functionality.

Event Modification

REQ-5.1 Managers must be able to create an event.

REQ-5.2 Managers must be able to include fields for the name, date/time and venue.

REQ-5.3 Managers must be able to delete an event.

REQ-5.4 Managers must be able to update an event's details.

REQ-5.5 Managers must be able to modify event branding (colour scheme, banner) to be displayed on the event's page for an event they have created.

Event Engagement

REQ-5.6 Members, or guests if the event is public-facing, must be able to subscribe (RSVP) to events.

REQ-5.7 Members or guests must be able to unsubscribe from events.

REQ-5.8 Members or guests must be able to receive reminders when a subscribed event is upcoming.

REQ-5.9 Members or guests must be able to receive updates when a subscribed event is changed.

REQ-5.10 Members or guests must be able to view a list of other members subscribed to an event.

REQ-5.11 Members or guests must be able to sort events based on their fields/data.

Event Moderation

REQ-5.12 Managers must be able to give members moderation permission for a particular event.

REQ-5.13 Moderators must be able to remove members or guests from events.

REQ-5.14 Managers must be able to view an audit log of changes to an event.

Content Submission

REQ-5.15 Managers must be able to enable content submission for an event.

REQ-5.16 Members or guests must be able to submit content to an event, which goes through an approval process as per [3.3 Knowledge Hub](#).

REQ-5.17 Members or guests must be able to view an event's submitted content, after it has been published.

3.6 Knowledge Graph

[\[Extension: Knowledge Graph\]](#)

REQ-6.1 Members are able to view various visualisations of the connections between members, tags found in their profile, and tags in Knowledge Hub entries.

3.7 Member Matching

[\[Extension: Content Synthesis\]](#)

Smart member matching uses AI to connect ASC members based on shared interests and skills. It processes member profiles, publications and activity to determine relevant suggestions.

REQ-7.1 Members must be able to see potential matches on a dedicated discovery page, based on keywords and concepts from member profiles and linked publications.

REQ-7.2 Members must be able to opt out of the matching system.

REQ-7.3 The system must dynamically update recommendations when a member updates their profile, adds a publication or interacts with new content.

REQ-7.4 The system must log member feedback on suggestions (e.g. viewed or dismissed suggestion) and use this information to improve future matching.

3.8 Content Synthesis

[\[Extension: Content Synthesis\]](#)

The content synthesis feature automatically generates monthly summaries on platform activity, trends, members and recent publications.

REQ-8.1 The system must generate a monthly summary on member activity, new members, recent publications and trending topics.

REQ-8.2 The system must distribute the summary to all members.

REQ-8.3 Members must be subscribed to the monthly summary by default.

REQ-8.4 Members must be able to unsubscribe from the monthly summary at any time.

REQ-8.5 Managers must be able to review and edit the summary before distribution.

REQ-8.6. Managers must be able to manually create and distribute a summary outside of schedule.

REQ-8.7 The system must analyse member activity across the platform to identify trends.

REQ-8.8 The system must show new members that have joined since the last summary.

3.9 AI-Powered Discovery

[\[Extension: Content Synthesis\]](#)

This feature uses AI to search external online sources for relevant papers, discussions and events. It delivers recommendations to members based on their profiles and platform activity.

REQ-9.1 The system must display personalised content on each member's feed.

REQ-9.2 The system must search external online sources based on member interests and platform activity.

REQ-9.3 The system must update a member's discovery feed when their profile is updated.

REQ-9.4 Members must be able to dismiss specific recommendations from their feed.

REQ-9.5 Members must be able to save or bookmark recommended content for later viewing.

REQ-9.6 The system must remember member feedback on recommendations for future discovery generation.

4. External Interface Requirements

4.1 User Interfaces

The system must function as a responsive web app, accessible by web browsers without the need for native installation.

[\[Extension: Platform Support\]](#) The system must be responsive, adapting layout to desktop, tablet and mobile screen sizes.

The system must support installation to a device home screen.

The system must provide consistent navigation for all user classes, with items shown or hidden based on Role Based Access Control.

The system must clearly display error messages, showing what went wrong and what action the user can take.

[\[Extension: Progressive Web App support\]](#) The system must be usable as a progressive web app.

[\[Extension: Progressive Web App support\]](#) [\[Extension: Events\]](#) the user's RSVPed events must be cached to be accessible offline.

The system must support the latest two versions of Chrome, Safari and Dia for all supported platforms.

[\[Extension: Platform Support\]](#) this extends to Firefox and Edge

4.2 Hardware Interfaces

The system must function on standard consumer desktop and laptop hardware. [\[Extension: Platform Support\]](#) This extends to laptops, tablets and smartphones.

[\[Extension: Platform Support\]](#) The system must support touch-based navigation on mobile devices.

4.3 Software Interfaces

The server must expose a REST API for communication between client and server.
The system must integrate with the ASC's Google Workspace mailing lists.

The system must **NOT** expose its database publicly, instead the back end will communicate with

the database locally.

The system must **NOT** directly handle payment information. It must instead use a payment processing provider to support any fees.

The system must support secure authentication with an authenticator-based OTP.

The system must support Argon2 for secure password hashing.

The system must expose an authentication server that other ASC applications are able to reuse.

4.4 Communications Interfaces

All client-server communication must exclusively occur over HTTPS.

The system must communicate with external academic sources over secure APIs.

The system must ensure HTTPS only communication with third party services, such as Anthropic's Claude and Stripe.

5. Other Nonfunctional Requirements

5.1 Performance

The app must display the main dashboard within 2 seconds after a successful login with valid credentials, under normal operating conditions (≥ 10 Mbps Connection).

The system must not take more than 1.5 seconds to load pages when the user is navigating through different components of the app under a load of up to 100 concurrent users.

The system must return search results within 2 seconds after a search request is performed by a user under a network speed of at least 10Mbps.

[\[Extension: Knowledge Graph\]](#) The system must render knowledge graphs containing up to 500 nodes within 5 seconds upon user request.

[\[Extension: Content Synthesis\]](#) The app must generate and display the profiles of researchers according to "smart member matching" within 3 seconds of the user navigating to the relevant dashboard.

[\[Extension: Content Synthesis\]](#) The app must regenerate the smart member matching components within 30 seconds if the user updates their profile or research interests. The smart scheduling feature must identify schedule conflicts of up to 50 invited people under 5 seconds of scheduling an event by a user.

The smart scheduling calendar must be updated within 5 seconds after an event is created by a

user.

[\[Extension: Content Synthesis\]](#) The monthly summaries must be displayed to the user within 1 second of clicking the notification.

[\[Extension: Content Synthesis\]](#) The app must display the AI-Powered Discovery within 3 seconds after the user navigates to the relevant dashboard.

5.2 Safety

The system must validate user inputs to protect the system from possible script attacks. The system database must automatically backup every 24 hours to prevent possible data loss.

The system must maintain a full audit log and version history for all collaboratively edited resources, and the administrators must be able to reverse any accidental or malicious modifications or deletions.

The system must display clear error messages during a system failure.

Critical information of users including event fees and payment methods stored in the system must be encrypted to ensure safety of users.

The administrators must have the ability to recover information and services within one hour after a critical system failure.

5.3 Security

The system must enforce strong password policies requiring a minimum of 8 characters including uppercase and lowercase letters, numbers, and special symbols in accordance with OWASP recommendations.

The system must use Argon2 password hashing algorithm combined with appropriate salting and cost factors to securely store user passwords.

The system must support Multi-Factor Authentication for all member accounts.

The system must implement JWT based authentication to manage user authentication sessions after a successful login.

The system must lock a user account if there are three consecutive failed login attempts to prevent brute-force attacks.

The system must implement Role-Based Access Control to restrict system access based on user roles, including member and administrator, ensuring users only have necessary permissions.

The system must ensure all personal and profile data is protected from unauthorised access.

The system must encrypt all sensitive data both in transit and at rest, using HTTPS/TLS.

The system must maintain audit logs of critical actions including logins, account changes, and administrative activities.

The system must protect against common web vulnerabilities including SQL Injections, Cross-Site Scripting and Cross-Site Request Forgery.

The system must store authentication token only in HttpOnly cookies to prevent Cross Site Scripting attacks.

The system must expire inactive user sessions after 10 minutes.

[\[Extension: Content Synthesis\]](#) The system must ensure that AI recommendations do not expose any sensitive research data.

The system must comply with GDPR and CCPA/CPRA privacy regulations.

5.4 Software Quality

The system must support at least 500 concurrent users without performance degradation.

The platform must maintain a minimum of 99.5% uptime annually, excluding scheduling maintenance, ensuring that it is available for users 24/7.

The system must ensure all content updates in the living cybernetic glossary are saved immediately and consistently.

The system must implement automatic error logging and recovery mechanisms to minimise service disruptions.

The system must provide an interactive and user-friendly interface that is suitable for users with varying levels of technical backgrounds.

The system must use a scalable architecture capable of supporting large knowledge graphs and increasing user loads.

The system must maintain a modular and testable codebase following standard software engineering practices.

The system must provide well-documented APIs and developer documentation to enable support for future development or extension and maintenance.

The system must log all AI and member activities, including edits and contributions, for required debugging and transparency.

Appendix A: Extensions

The following features are not part of the Minimum Viable Product. They are presented in decreasing order of priority.

Knowledge Graph

[Section 3.6](#) and related features.

Events

Section [3.5](#) and related features

Content Synthesis

Sections [3.7](#), [3.8](#) and [3.9](#) and related features.

Progressive Web App support

The ability to use the CKH as a “progressive web app”.

Platform Support

Support for accessing the CKH from linux devices

Support for accessing the CKH from smartphones and tablets.

Support for accessing the CKH from firefox and edge.

Network Analysis Workshop

Network Analysis Workshop

Potential — meeting early June ([When2Meet](#))

Expert: Valdis Krebs, Network Scientist, LinkedIn [profile](#), [website](#)

Network analysis, network visualization, organizational networks, community networks

Introduction sent to him

A student team based in Perth, Australia is developing a knowledge hub for the American Society for Cybernetics (ASC) — a node-network interface that would allow members to navigate connections between people, projects, book references, and ideas in an *exploratory* rather than hierarchical way. Their **challenge** is a real one: **how do you design for *discoverability* and *emergence* in a knowledge network, rather than defaulting to passive consumption?**

They've been asking: who in the world thinks most rigorously about network structure and meaning? Your name came up immediately.

What we are proposing is not a lecture. Here's the format:

- **Before the session**, students would watch a short existing video or read a brief paper of your choosing — something that gives them a working vocabulary for the conversation.
- They would then **submit one or two paragraphs framing their challenge** (not as a description of their app, but as a genuine design question).
- **Session**: We would meet for approximately one hour — you would respond to their challenge, probe their assumptions, and guide the group toward approaches and tradeoffs worth considering.

The idea is that you encounter a real problem situation and demonstrate your thinking in dialogue. The students learn through the exchange; you (hopefully) enjoy the puzzle.

This would be unpaid, though we recognise that is a significant ask of your time. We are hoping the combination of an interesting challenge, a genuinely curious group of young cybernetics students, and a light-touch format makes it feel worth an hour.

Would you be open to this?

Articles

Articles — Node Graphs

1 — A recent blog set of 3 posts is popular now ... all about distance in networks. Sounds like that will be a dynamic they want to explore. (copied below)

2 — Since we are looking beyond a purely human network (including ideas, books, projects, events) we probably want to look at two/multi-mode networks. Here is how books form networks based on human/reader's interests/likes... [\(PDF\) Proxy Networks](#)

3 — Two-mode networks ... also covers book networks... [Your Choices Reveal Who You Are: Mining and Visualizing Social Patterns](#) (PDF)

PDFs copied to [gDrive Articles folder](#)

Distance in Human Networks

Part 1

<https://www.rsu.lv/en/news/distance-human-networks-part-one>

23 January 2026

With this blog post, we begin a new three-part series by network analysis expert and (R)E-TIES researcher Valdis Krebs on distance in human networks. In the first post, the researcher explores how distance in human networks limits influence and information flow beyond a three-step horizon while increasing message distortion due to contextual differences.

When we join a network, we think we can quickly get anywhere in the network and access all the information and knowledge we need, no matter where it is in the network.

Look at the network below in Figure 1. It is an actual network of a small technology business. Assume you are the red node on the left side of the diagram. Your direct, one-step contacts are the yellow nodes. These are the people you mostly work with to get your job done. Your two-step contacts are the green nodes, which are connected to the yellow nodes. These people influence your work and provide information and knowledge that is not immediately available inside your work group. The blue nodes are three steps away and they are only connected to the green nodes. These people probably do not influence how the red node accomplishes their daily tasks. They only influence the red node rarely and indirectly.

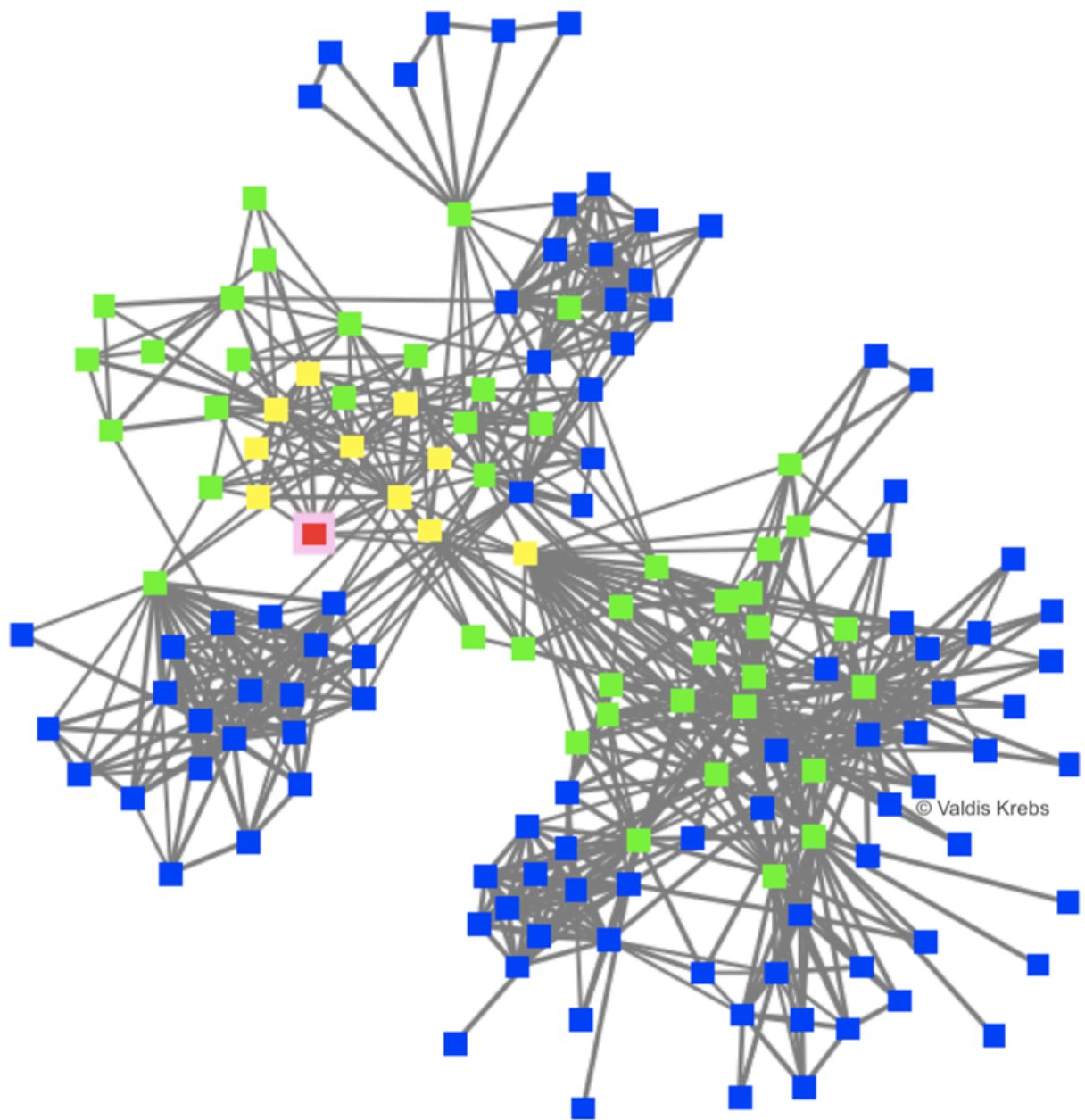


Figure 1. Frequent Symmetric Work Ties in a Technology Business

What we notice in networks is that they all have a horizon - the further someone is away from us the harder they are to see and the less useful they are to our daily needs.

Just as we have geographical horizons, we also have distances beyond which we cannot see into our network. Based on the research of Friedkin, Granovetter, and Christakis & Fowler, a network horizon does not go past 3 steps, or 2 intermediaries. Whether you are looking for job,

searching for information, trying to find an expert, or aiming to influence others, the limit appears to be 3 steps in your network. Beyond that you are blind and powerless.

Another problem with distance in networks is distortion. As a message travels from person to person it becomes distorted. We all knew this as kids when we played the “Telephone Game”. As a message is told and retold, in a chain of people, it soon becomes unrecognizable by the original teller.

Context between teller and listener of a message affects message distortion. The more two people share the same context (and therefore use the same lingo and have the same history) the better they can communicate and keep the distortion minimized. Figure 2 shows how a message slowly gets distorted with each re-telling. We see how three different contextual environments affect the outcome of the message chain. The percentage number, in Figure 2, between two nodes indicates how much of the message is passed on. In the end we multiply all the percentages to see how much of the original message survives. People telling a message to those they have little in common with soon kill the whole meaning of the message.

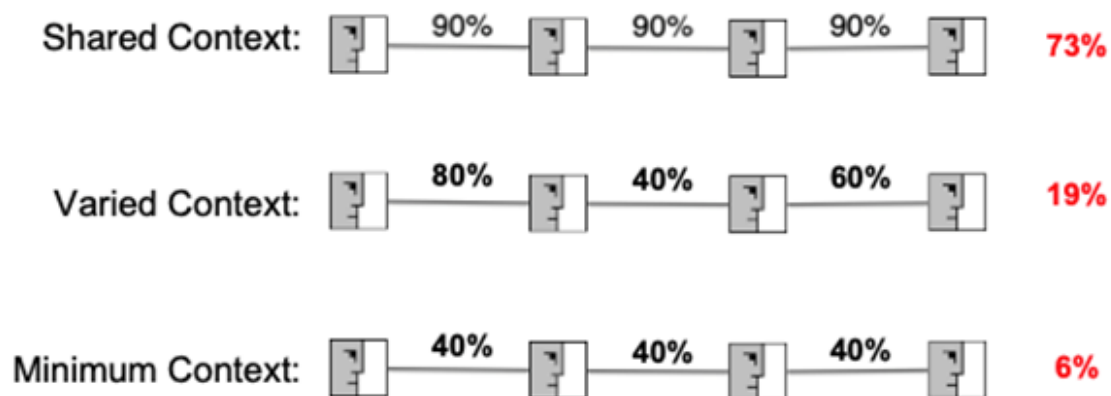


Figure 2. Distortion of communication within and between contexts

Some might say that a message does not distort if everyone gets the same message at once like the same email sent to many. That way, the reasoning goes, everyone reads the same content and no re-telling chain is required for distribution. Direct access - no distortion! No. Again context comes into play to influence the meaning of the message. If two people don't share the same context, nor have the same history of an ongoing conversation or project they will not interpret the same words the same way.

Each department in an organization has their definitions and internal lingo. I once worked in a large organization where the Human Resources, Employee Benefits, and Finance departments could not agree on a simple term: “employee headcount”. They each had their own rules and came up with 3 different numbers for total headcount. In the end, who was right? They all were. Each department was working under its own rules, policies, procedures and laws. It turns out that “an employee” can be defined in multiple ways with multiple types of employees. These

three departments work closely together. Can you imagine the possible misunderstandings in communication between departments that interact rarely and don't know each other well? Distance can be dangerous - to understanding and agreement.

Part 2

With this blog post, we continue the three-part series by network analysis expert and (R)E-TIES researcher Valdis Krebs on distance in human networks. In this installment, the researcher discusses the benefits of some distance in human networks.

In part one of this blog series, we discussed some drawbacks of distance. But there are also benefits to some distances. When we collaborate to do our work, our direct colleagues are key to our productivity. We cannot work alone, or know everything ourselves, so we keep knowledge and trust with those closest to us. When looking at getting things done it is important to look at the “network neighborhood” around an employee - those connections that are one step and sometimes two steps away. A person embedded in a good network neighborhood will be more successful. Unfortunately, the reverse is also true, hire a superstar and put them in a bad network neighborhood and they will struggle. This is obvious to most people - surround yourself with good people and you will be fine. Figure 3 shows the relationship between network distance and mostly direct ties in collaboration (1 step) it is your inner circle that matters most.

What is not so obvious is how distance aids innovation. The people around us help us get things done, but they are not so useful for when we need to do things differently - come up with new ideas. They mostly know what we know, at about the same time. Innovation requires new ideas and different thinkings, but not too different. Here again context is key. We need people who mostly share our context but reside in nearby locations in the network. Those in adjacent neighborhoods will have different ideas, hear alternate approaches and opinions and will have different histories to the similar topics.

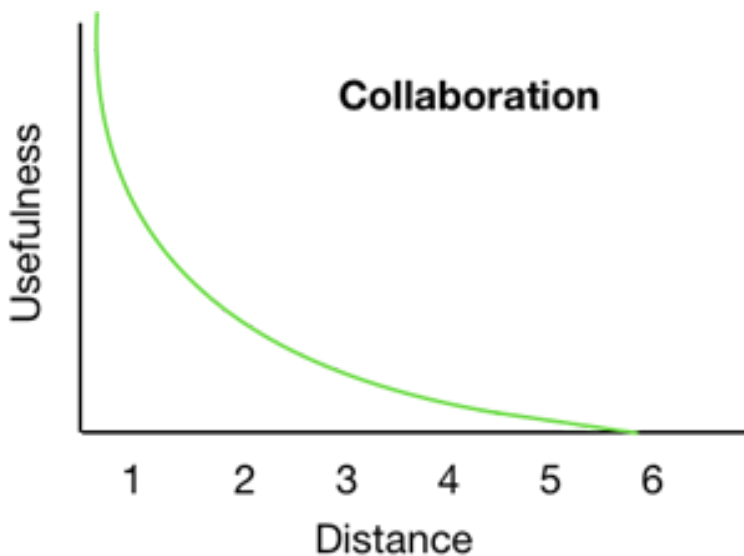


Figure 3. Distance vs. Usefulness for Collaboration

The benefit of adjacency is that these neighbors are different, but not too different. You have much in common and can talk a similar language and share similar experiences. As I tell my clients and students: “Connect on Your Similarities, and Benefit from your Differences”.

These network neighbors are within your network horizon, so you can see them, watch them, learn, and maybe be introduced to them at the right time by a friend or colleague in common. The sweet spot for innovation appears to be those who are currently two to three steps away from you. When looking for opportunities do not look in your back yard, visit adjacent neighborhoods. Figure 4 shows where the “just right” Goldilocks solutions are in one’s network.

It appears that in human networks it is the one, two, and three step paths that matter. Although in real human networks there are many paths that are longer than three. What flows on these paths? Is it noise? Or maybe incomplete flows? A message starts at one end of a nine step path and starts moving forward, but at the fourth or fifth step it reaches a context boundary and the message never gets across, it is dropped. So, the nine-step path is there but things never flow the whole distance they just utilize the path for shorter trips? Get on at step 1 and get off at step 3, get on at step 5 and get off at step 8, and so on.

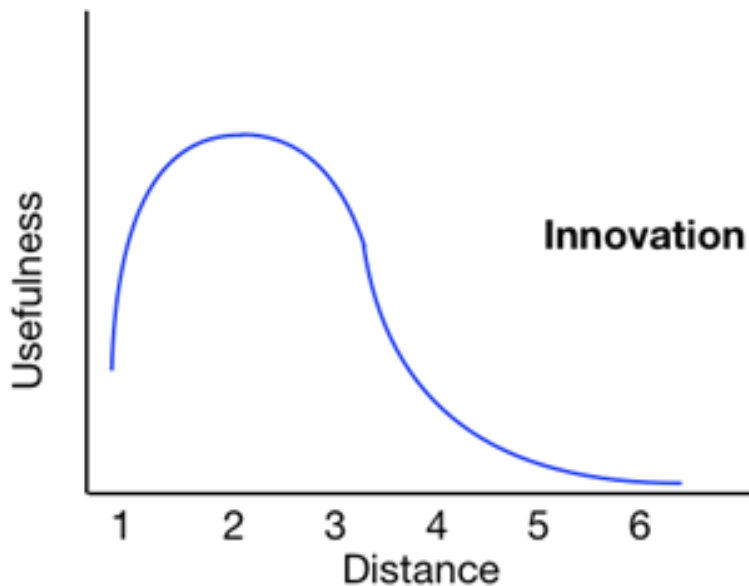


Figure 4. Distance vs. Usefulness for Innovation

These are all shorter paths that are utilized and should be paid attention to. The nine-step paths might just be background structures that are never fully utilized? Maybe we look at the nine-step path as several overlapping shorter paths that are utilized. We saw how quickly messages get distorted through just three re-tells. Can you imagine the distortion of the message after 9 relays?

We will investigate how what we have learned affects network metrics in part three.

Part 3

With this blog post, we conclude a three-part series by network analysis expert and (R)E-TIES researcher Valdis Krebs on distance in human networks. In this final installment, the researcher argues that traditional network metrics relying on geodesics (shortest paths) can distort how communication actually flows, and proposes alternative metrics based on shorter, more frequently used paths for more accurate analysis.

Current human network metrics are calculated using geodesics - the shortest path between any two nodes in a network, whether that path is 2 steps or 12 steps. All geodesics are calculated and then utilized in the network metrics.

We saw from our distortion example that information that travels on a long network paths are both distorted and delayed and might be dropped from the communication path along the way. Why include those long paths in a calculation? We also know that communication happens on other short paths, not just via geodesics. Figure 5 shows a typical communication example that does not use an available geodesic - it uses a longer, yet more frequent path.

Even though the moms in Figure 5 have a direct link (the geodesic), most of the information flow between them is via their children, who are friends and spend a lot of time together every day at school. The thicker links represents more frequent communication. Rather than the 1 step geodesic being utilized (the shortest path), a more frequently used 3-step path is utilized: mom A - child A - child B - mom B.

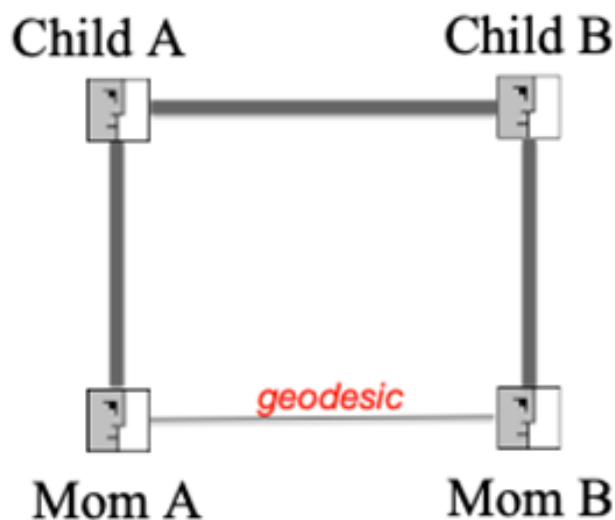


Figure 5. Communication Paths: Actual vs. Shortest

The moms and their children communicate daily, as do the children at school. The longer, more frequent, path is used instead of the less frequent direct path.

This same pattern of using the more efficient but longer path, instead of the shorter path, is also seen in offices with employees and supervisors. Employees from different departments, working on the same project, communicate more frequently, than their supervisors who may only meet as needed.

Seeing the futility of long paths in a network I developed a new set of network metrics that utilize all short paths (one, two and three steps) in the network. We look at all paths between two nodes, not just geodesics, as long as those path lengths are less than four steps (not beyond the network horizon).

Figure 6 shows a client organization where we calculate all the geodesics in the network. We see that some employees are very far apart when it comes to sharing work information. Two manufacturing employees working on two different products, on two different machines, for two different customers, at two different locations probably have no relevant work information to share.

So why include 1.75 million geodesics > 3 steps in the network calculation? Do they provide greater accuracy? Or do those long geodesics distort who is central in the network? It is most likely the later (This approach works for human networks and any other networks with flow where distance distorts and delays and maybe drops the signal).

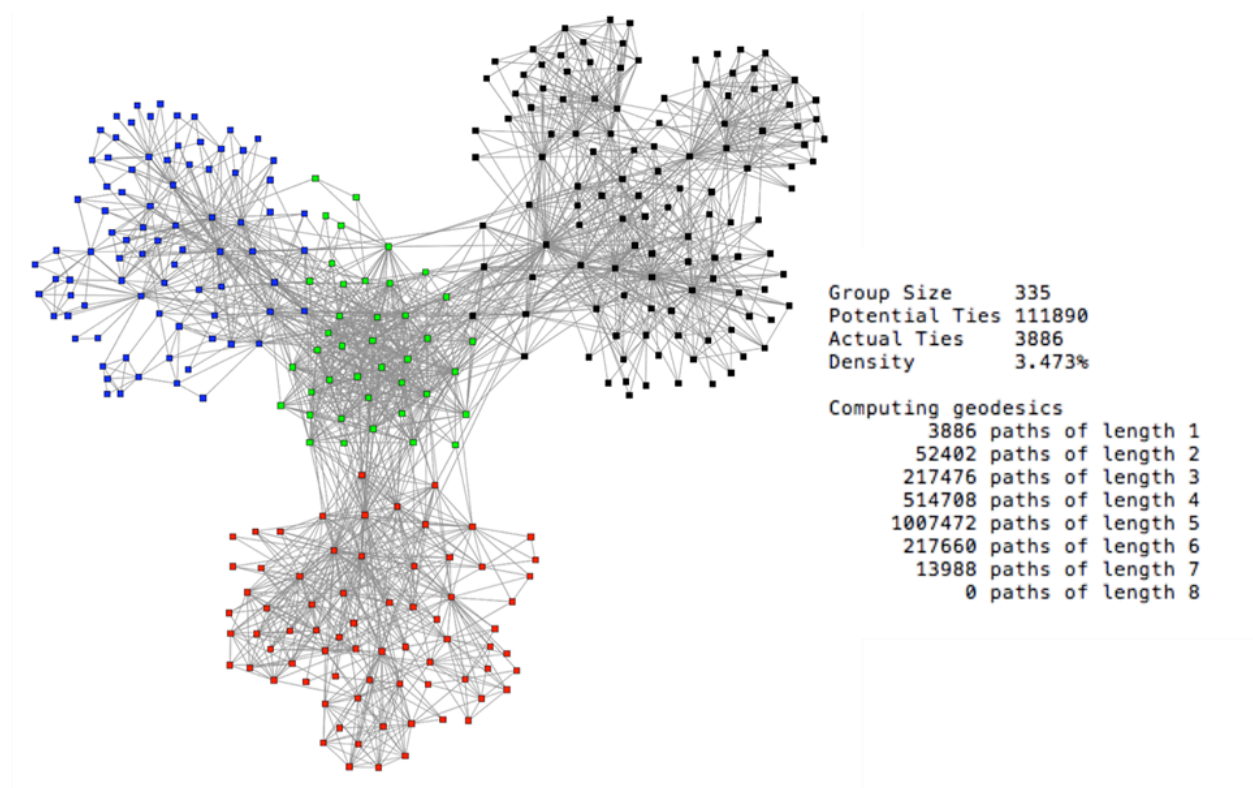


Figure 6. Complete Network of Work Ties in Manufacturing Business

Let's compare our non-geodesic metrics to commonly used network centralities. First let's look at the popular Betweenness metric by Freeman (1979). It measures which nodes have the most

geodesics flowing through them - revealing who controls the flow? We compare it to our Connector metric - which nodes have the most 2 and 3 step paths flowing through them?

Betweenness looks at the whole network graph, no matter how long the paths. Connector just looks within the network horizon of each node. Figure 7 shows us that both metrics return highly similar results. The size of the node in each network shows the value of that network metric.

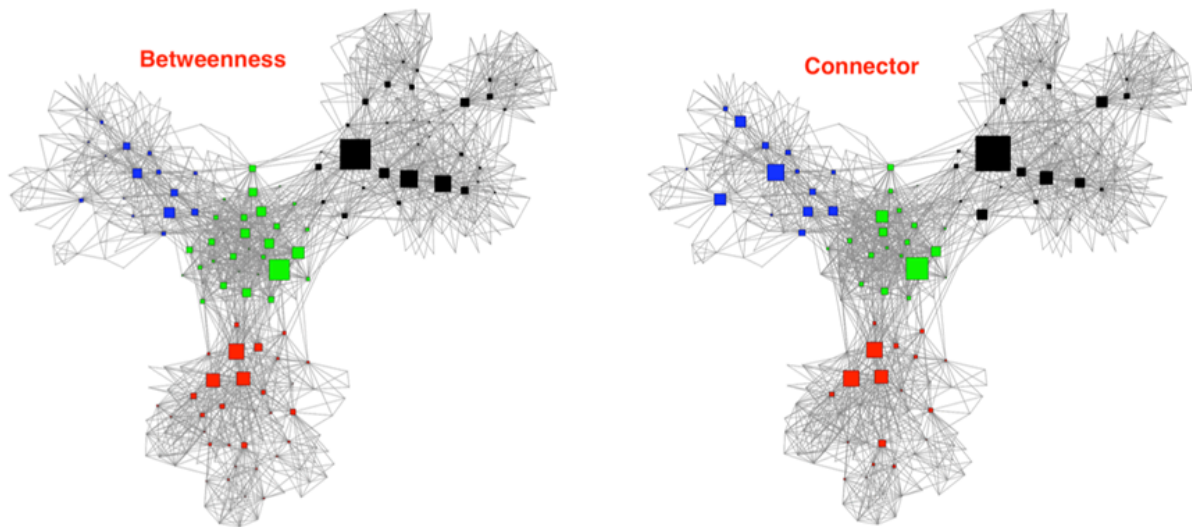


Figure 7. Comparing Network Metrics: Integration vs. Eigenvector Centrality

Figure 8 shows us how the popular Eigenvector centrality compares to our Integration metric. Both metrics are based on the notion that it is better to be connected to others who are well connected. Nodes in the diagram are sized according to the value of the stated centrality metric. Both metrics choose highly similar nodes. But the Integration metric seems to distinguish the centralities better looking at the range of node sizes in each network.

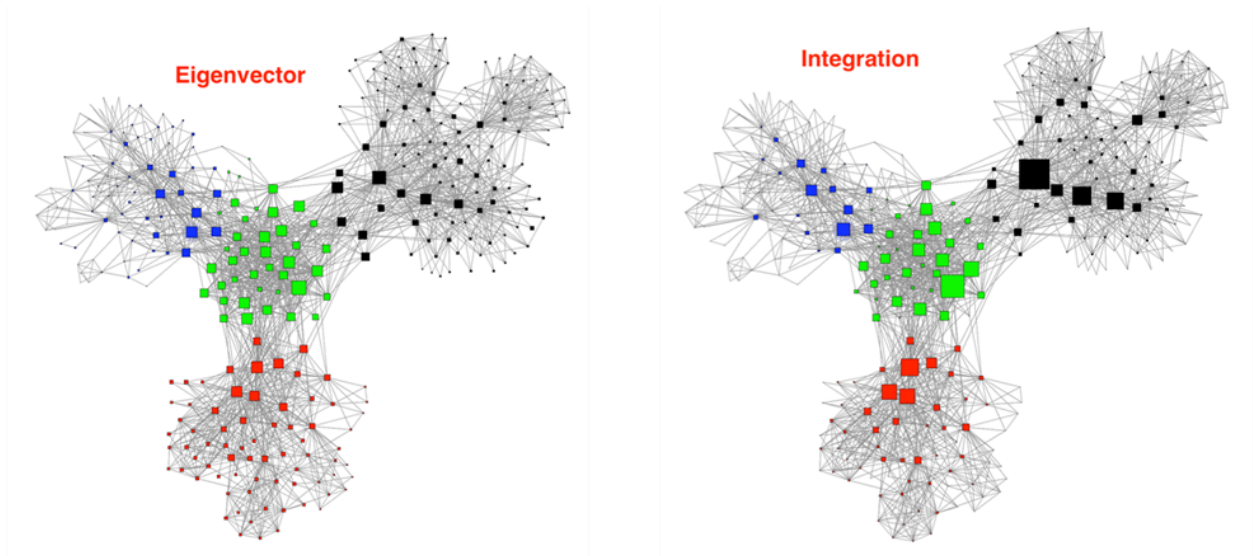


Figure 8. Comparing Network Metrics: Integration vs. Eigenvector Centrality

Besides saving needless compute time by not calculating millions of geodesics, the algorithm and names of these new metrics are an advantage. We can easily explain them to a normal client. People are more likely to accept what they understand.